

WTO Accession, PNTR, and U.S. Import Quantities and Prices

Vincent Zheng, Felix Yu, Aryan Goel

March 3, 2026

Outline

- Background and research question
- Data and empirical strategy
- Current results and robustness
- Next steps

Background

- China's economy grew rapidly in the 1990s, becoming a major exporter to the U.S.
- Under U.S. law, China received low tariff rates (Normal Trade Relations) but Congress had to **vote to renew** NTR status every year.
- This annual renewal created **trade-policy uncertainty**
 - importers faced the risk that tariffs could spike to 1930s-era levels if Congress voted no.
- In **October 2000**, the U.S. passed the law granting China Permanent Normal Trade Relations (PNTR), conditional on WTO accession.
- In **December 2001**, China officially joined the WTO; permanent low tariffs took effect in January 2002.
- We take advantage of the fact that the actual tariff rates mostly **did not change**: what changed was the **elimination of uncertainty** about future tariffs.

Research question

How did the resolution of U.S. trade-policy uncertainty around China starting in 2002 affect:

- the volume of imports from China into the U.S., and
- the prices U.S. buyers paid for Chinese imports?

Policy Timing and Identification Intuition

- **Pre-2001:** China receives NTR annually; risk of non-renewal creates uncertainty that is priced into trade decisions.
- **2000:** U.S. grants PNTR (legal permanence, conditional on WTO accession).
- **Dec 2001 / Jan 2002:** China joins WTO; permanent low tariffs take effect.

Why did uncertainty matter differently across products?

- Every product had two legislated tariff rates: a low NTR rate and a high non-NTR rate (dating back to the Smoot-Hawley Act of 1930).
- The **NTR gap** = non-NTR rate – NTR rate measures how much a product's tariff *would have jumped* if Congress didn't renew.
- Products with large gaps faced more risk → importers could be heavily impacted.
- Products with small gaps faced less risk → less impact by the annual vote.

Identification Strategy: Difference-in-Differences

Core idea

Compare import growth for **high-gap** products (more exposed to uncertainty) vs. **low-gap** products (less exposed), before and after WTO entry.

Why this works:

- Low-gap products serve as a **control group**: because, while they were still affected, they had a small stake from the policy change.
- Any “extra” import growth in high-gap products after 2002, beyond what low-gap products experienced, can be attributed to the removal of tariff uncertainty.

Treatment exposure (measured in 1999, before WTO negotiations):

$$\text{Exposure}_p = \text{gap_pre}_p = (\text{non-NTR tariff})_p - (\text{NTR tariff})_p$$

Products above the median gap $\rightarrow$ “high-gap” (treatment); below $\rightarrow$ “low-gap” (control).

Data

Data sources

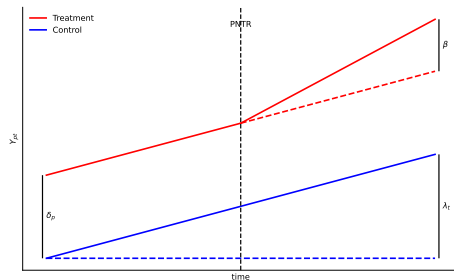
- USITC monthly imports by product and HTS.
- Tariff schedule information used to construct product-level `gap_pre`.

Baseline Specification

Two-way FE DiD

$$Y_{pt} = \beta (\text{HighGap}_p \times \text{Post}_t) + \delta_p + \lambda_t + \varepsilon_{pt}$$

- Y_{pt} : log import value, log quantity, or log unit value.
- δ_p : product fixed effects.
- λ_t : time fixed effects.
- ε_{pt} : error term.



Main DiD Results (Current Outputs)

Binary treatment (HighGap x Post) with product and date FE

- `ln_customs_value`: $\hat{\beta} \approx 0.121$ (s.e. 0.031), positive and significant.
- `ln_first_qty`: $\hat{\beta} \approx 0.145$ (s.e. 0.036), positive and significant.
- `ln_unit_value`: $\hat{\beta} \approx -0.024$ (s.e. 0.017), negative but not strongly significant in FE spec.

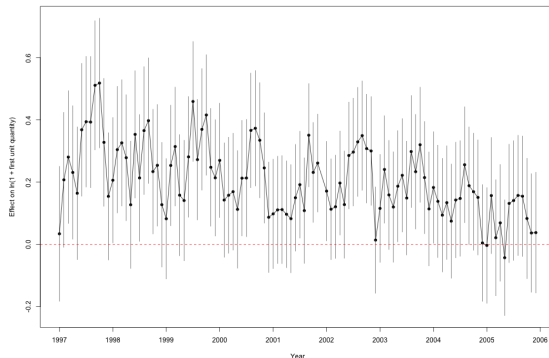
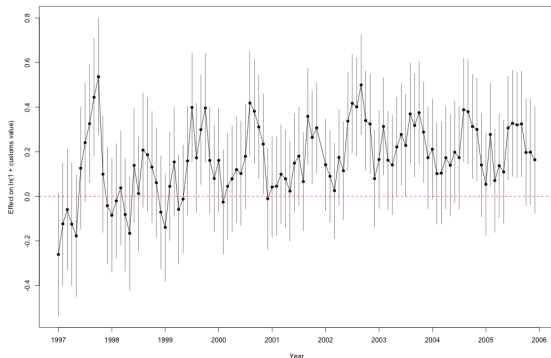
Interpretation

- High-gap products show stronger post-policy growth in value and quantity.
- Price effects are weaker in the binary split and clearer in continuous treatment.

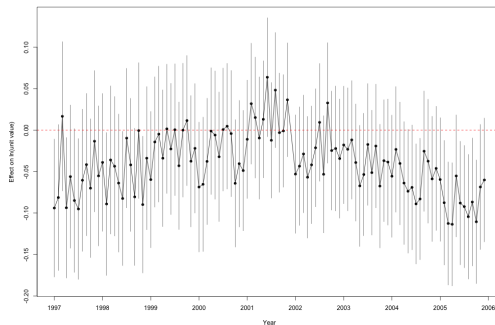
Event Study: Value and Quantity

Event-study specification

$$Y_{pt} = \sum_{k \neq -1} \beta_k (\text{HighGap}_p \times \mathbf{1}[\text{event_time}_{pt} = k]) + \delta_p + \lambda_t + \varepsilon_{pt}$$



Event Study: Unit Value (Price Proxy)



Continuous-treatment pre-trend test for unit value: $p \approx 0.081$ (marginal).

Takeaways and Next Steps

Takeaways

- Evidence supports a post-policy increase in import value and quantity for high-gap products.

Next steps

- empty

Questions